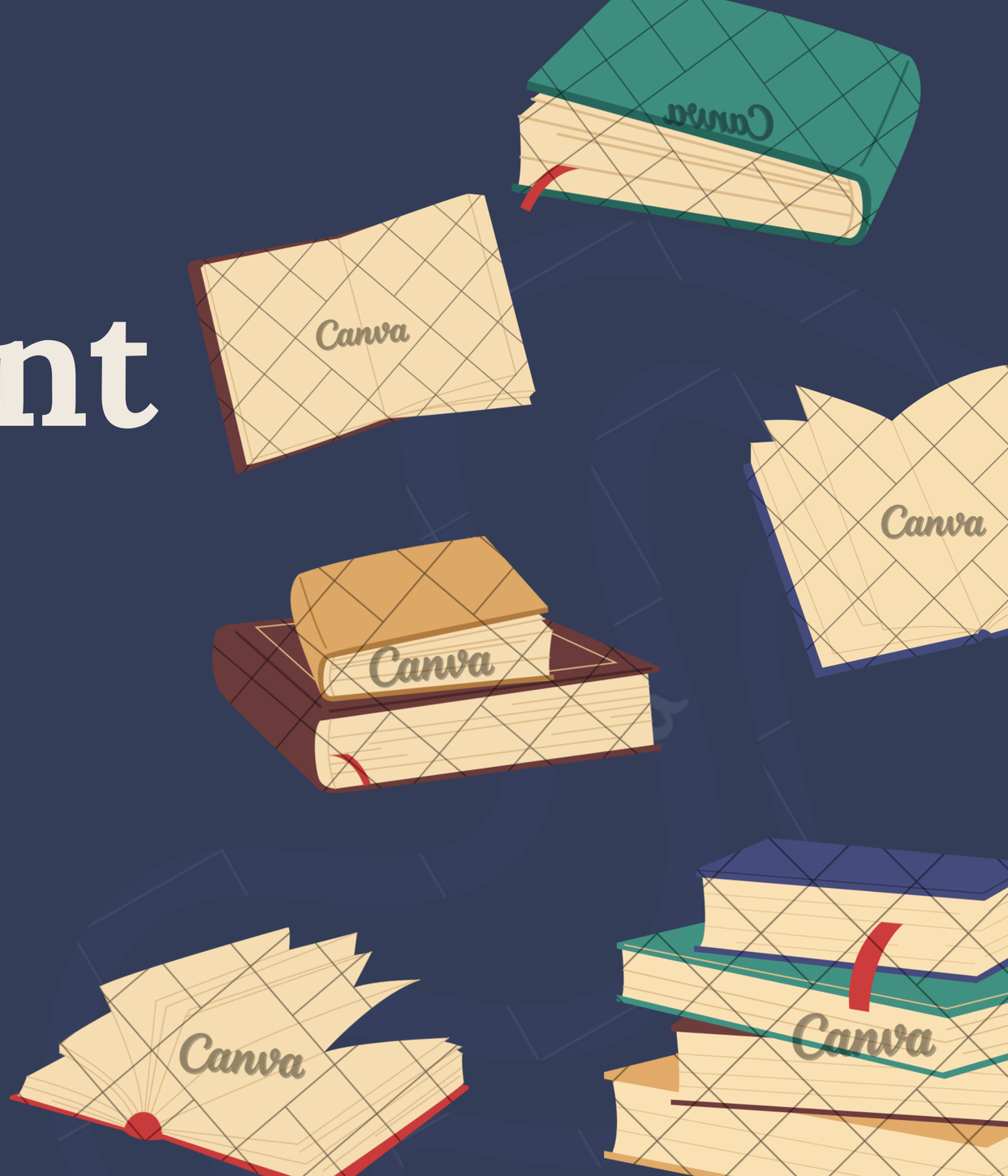
Library Management System

Ethan, Andrew, Mauricio



Our Idea

- Traditional library systems are often outdated and difficult to navigate.
- Students need efficient ways to find, reserve, and manage borrowed books.
- Librarians need efficient tools for inventory management.
- Our project is useful to any library that has not already upgraded to a digital system.

Context

- Use Cases:

- A student searches for a book on "database design" and places a hold if it's checked out.
- A librarian scans a book barcode to check it out to a student.
- The system automatically calculates and applies fines for overdue books.
- An administrator generates a report of most borrowed books this semester

Target Users

- Students
- Faculty
- Librarians
- Administrators

Students / Faculty

- Advanced search (by title, author, ISBN, subject, genre)
- View book availability and location in library.
- View borrowing history and current loans.
- Renew materials online.
- Place holds or reservations on checked out items.
- View and pay fines.
- Save favorite books/reading lists.

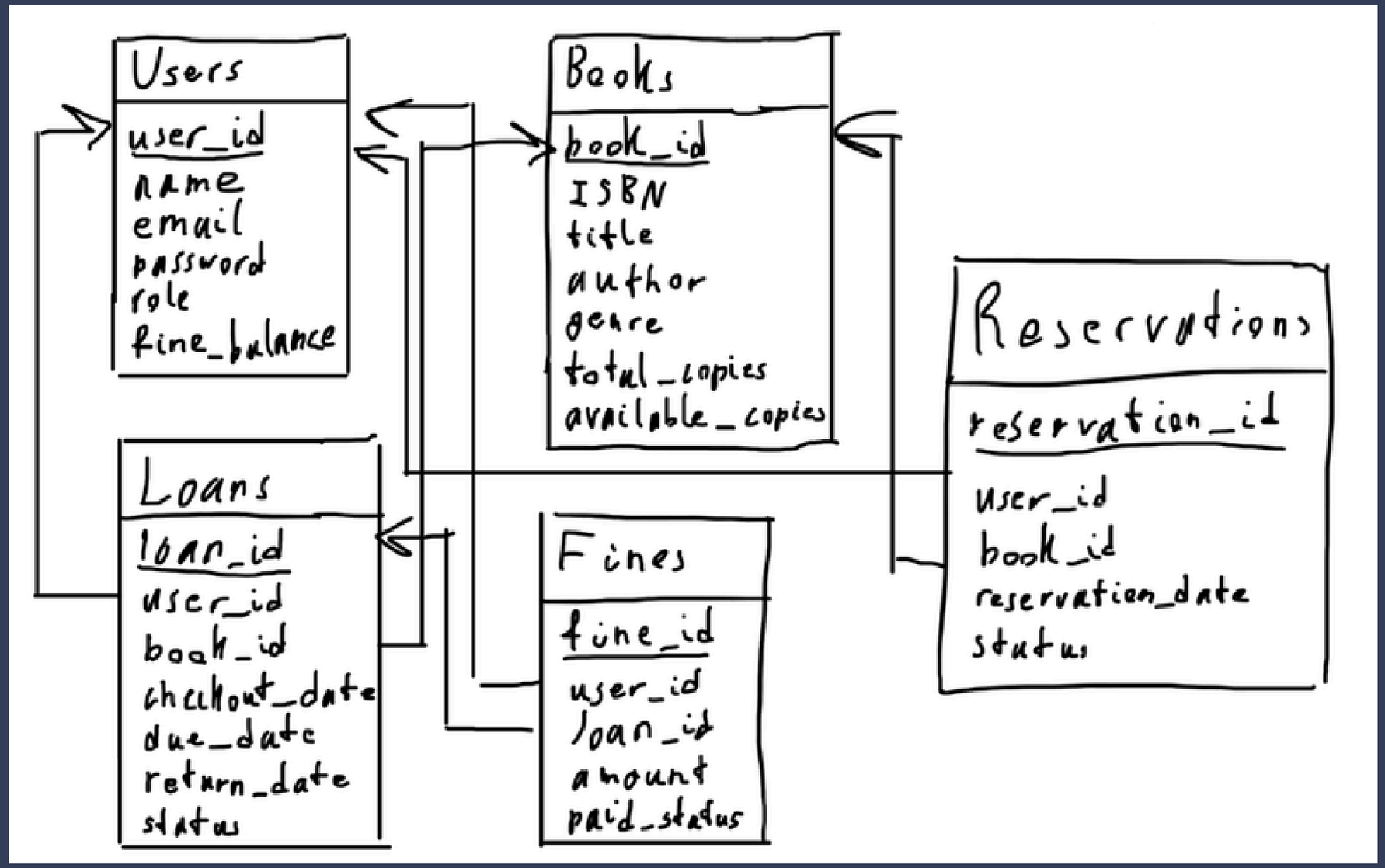
Librarians

- Process checkouts and returns.
- Add, edit, and remove books from catalog.
- Manage user accounts.
- Override due dates or fines when needed.
- View reservation queue for popular items.
- Generate reports.

Administrators

- Comprehensive analytics dashboard.
- User management and role assignment.
- Configure loan periods and fine rates.
- Inventory reports (missing books, damaged items)
- Popular books and usage statistics.

Database Structure



Task Distribution

Ethan: Frontend
Andrew: Backend
Mauricio: Database

Questions?